A NUMERICAL METHOD FOR SOLVING THE EIKONAL EQAUATION ON SIMPLICIAL TESSELATIONS*

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Abstract. In this paper, we describe a method for solving the Eikonal equation on domains in $\mathbb{R}^2$ or $\mathbb{R}^3$ tesselated by triangles or tetrahedra. Our method generalizes to simplices in higher dimensions, owing to our choice of solution representation under the Jacobi family of orthogonal polynomials on the simplex. First, we describe a spectral decomposition for functions supported on the simplex in terms of their weighted L_2 expansion coefficients under the Jacobi basis. Computing this expansion requires an accurate and efficient quadrature, and we provide a novel optimization routine based on approximate joint-diagonalization of Jacobi matrices for generating such quadratures. Then, we consider differential operators acting on coefficients of a Jacobi expansion. Recent results due to Townsend, Oliver and Vasil enable the construction of sparse differential operators on the standard triangle, interoperating within the Jacobi polynomial family parameter space, and leading to low-storage, banded discrete differential operators. We generalize these results to the tetrahedra (laying the path for generalizing to $d > 3$), and use the resulting differential operators to construct the Eikonal operator on the simplex. Ultimately, we must solve a non-linear PDE, and working in coefficient space implies that our solution variables must be thought of as functions. Hence, we reformulate the Eikonal problem in terms of an infinite-dimensional L_2 mininization problem, which can be discretized with the aforementioned numerical quadrature. As with the README, this is more of a blog, but it's much easier to typeset math here.

Key words. multivariate orthogonal polynomials, spectral methods, partial differential equations, Eikonal equations, simplex

MSC codes. 68Q25, 68R10, 68U05

1. Global representations to local solutions of the Eikonal Equation.

Haven't really looked into the viscosity perspective, but..I think I converged on the same idea for solving the PDE. Consider the standard simplex

$$\mathcal{T}^d = \{\boldsymbol{x} = (x_1, \cdots x_d) \in \mathbb{R}^d \mid x_i \geq 0,\ 1 - |\boldsymbol{x}|_1 \geq 0\}$$

The *minimal* time $u(\boldsymbol{x})$ to reach the boundary $\partial\mathcal{T}^d$ from a point $x \in \mathcal{T}^d$ is computable through solving the Eikonal equation on $\mathcal{T}^d$:

$$\begin{cases} \sum_{i=1}^d (\partial_{x_i} u)^2 = \left(\frac{1}{f}\right)^2, & x \in \mathcal{T}^d, \\ u = q, \quad x \in \partial\mathcal{T}^d, \end{cases} \tag{1.1}$$

where $f : \overline{\mathcal{T}^d} \to \mathbb{R}^+_\star$ is the speed in the medium, $q : \partial\mathcal{T}^d \to \mathbb{R}^+$ is the time to leave $\mathcal{T}^d$ from a point on $\partial\mathcal{T}^d$. We seek to represent u under an optimal, finite, weighted L_2 expansion, wherein

$$u(\boldsymbol{x}) \approx \sum_{j=1}^{n} u_j P_j(\boldsymbol{x}) = \boldsymbol{u} \cdot \boldsymbol{P}(\boldsymbol{x}),$$

and $\boldsymbol{u}$ is the vector in $\mathbb{R}^n$ of coefficients of u under the elements of the basis $\boldsymbol{P}(\boldsymbol{x}) : \mathbb{R}^d \to \mathbb{R}^n$. Here $\approx$ must be taken in the sense of weighted L_2 convergence of the series to u as $n \to \infty$. For the typical case of $q \equiv 0$ (i.e. time to boundary is 0 if on the boundary), we employ a weighted basis which evaluates to 0 on the boundary, thereby naturally satisfying homogeneous Dirichlet conditions.

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1.1. Jacobi bases on the Triangle. Let us specialize for $d = 2$. By P_k^n, we denote the k^{th} Jacobi polynomial on the triangle of degree n with parameters (a, b, c). When the parameters must be specified, we will refer to the same polynomial by $P_{k,n}^{(a,b,c)}$. The weight function is

$$\begin{aligned} W^{(a,b,c)}(x,y) &= x^{a-\frac{1}{2}}y^{b-\frac{1}{2}}(1-x-y)^{c-\frac{1}{2}}, \\ w_{(a,b,c)} &= \Big(\int_{\mathcal{T}^2_{\text{ref}}} W(x,y)dxdy\Big)^{-1} = \frac{\Gamma(|\kappa|+\frac{3}{2})}{\Gamma(a+\frac{1}{2})\Gamma(b+\frac{1}{2})\Gamma(c+\frac{1}{2})} \\ w^{(a,b,c)}(\boldsymbol{x}) &= w_{(a,b,c)}W^{(a,b,c)}(\boldsymbol{x}) \end{aligned} \tag{1.2}$$

with $a, b, c > -1/2$ the Jacobi parameters, and $|\kappa| = a + b + c$. The orthonormal polynomials are given in terms of the 1D Jacobi polynomials by

$$\begin{aligned} P_k^n(x,y) &= (h_{k,n})^{-1}P_{n-k}^{2k+b+c,a-1/2}(2x-1)(1-x)^kP_k^{c-1/2,b-1/2}\Big(\frac{2y}{1-x}-1\Big), \\ (h_{k,n})^2 &= \frac{w_{a,b,c}}{(2n+|\kappa|+\frac{1}{2})(2k+b+c)} \\ &\quad\times\frac{\Gamma(n+k+b+c+1)\Gamma(n-k+a+\frac{1}{2})\Gamma(k+b+\frac{1}{2})\Gamma(k+c+\frac{1}{2})}{(n-k)!k!\Gamma(n+k+|\kappa|+\frac{1}{2})\Gamma(k+b+c)}, \end{aligned} \tag{1.3}$$

and they satisfy a mutual orthonormality condition under the weighted inner product:

$$\langle P_{j,m}^{(a,b,c)}, P_{k,n}^{(a,b,c)}\rangle_{w^{(a,b,c)}} = \delta_{jk}\delta_{mn}. \tag{1.4}$$

A given basis polynomial is not only orthonormal to those with different total degree, but also to those others within the same-degree homogeneous subspace. In other words, P_k^n, $0 \le k \le n$ forms a basis for the space of homogeneous polynomials of degree n. By considering ordered collections of these subspaces (where ordering is in terms of the degree n of the space), instead of individual polynomials, we can more elegantly reformulate the Eikonal problem into a weak-type form. Let

$$\mathbb{P}_n(x_1,x_2) = (P_0^n(x_1,x_2),\cdots P_n^n(x_1,x_2))^T,$$

and let

$$\boldsymbol{P} = (\mathbb{P}_0^T,\mathbb{P}_1^T,\cdots,\mathbb{P}_{n-1}^T)^T.$$

The above notation amounts to a vectorization of the basis for the general space of polynomials in two variables Π^2_{n-1} defined of $\mathcal{T}^2$.

Now, consider the $(n-1)$ order Jacobi polynomial expansion of the solution u under the parameter family (a, b, c). For $\boldsymbol{x} \in \mathcal{T}^d$, and $N = \dim\Pi^2_{n-1}$ coefficients $\boldsymbol{u} \in \mathbb{R}^N$, we can write it as

$$u(\boldsymbol{x}) = \boldsymbol{P}^{(a,b,c)}(\boldsymbol{x})^T\boldsymbol{u}. \tag{1.5}$$

There exists sparse discrete operators which map $\boldsymbol{u}$ to the coefficients of the derivative of u under a modified basis, as well as those which promote the basis parameters without taking derivatives:

$$\begin{aligned} \partial_{x_1}u(\boldsymbol{x}) &= \boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T\mathcal{K}^{(a+1,b+1,c+1)}_{(a+1,b,c+1)}\mathcal{D}_{x_1}\boldsymbol{u} \\ \partial_{x_2}u(\boldsymbol{y}) &= \boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T\mathcal{K}^{(a+1,b+1,c+1)}_{(a,b+1,c+1)}\mathcal{D}_{x_2}\boldsymbol{u} \end{aligned} \tag{1.6}$$

Inserting this into the fully constrained Eikonal equation gives

$$\begin{aligned}
&\left(\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T \mathcal{K}_{(a,b,c)}^{(a+1,b+1,c+1)} \boldsymbol{c}_{\frac{1}{f}}\right)^2 = \quad (1.7)\\
&\left(\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T \mathcal{K}_{(a+1,b,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_1} \boldsymbol{u}\right)^2 + \\
&\left(\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T \mathcal{K}_{(a,b+1,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_2} \boldsymbol{u}\right)^2 = \\
&\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T \left(\tilde{\mathcal{D}}_{x_1} \boldsymbol{u}\boldsymbol{u}^T \tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2} \boldsymbol{u}\boldsymbol{u}^T \tilde{\mathcal{D}}_{x_2}^T\right) \boldsymbol{P}^{(a+1,b,c+1)}(\boldsymbol{x}), \quad (1.8)
\end{aligned}$$

where

$$\tilde{\mathcal{D}}_{x_1} = \mathcal{K}_{(a+1,b,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_1}, \quad \tilde{\mathcal{D}}_{x_2} = \mathcal{K}_{(a,b+1,c+1)}^{(a+1,b+1,c+1)} \mathcal{D}_{x_2}$$

and, the coefficients of $(1/f)$ are represented as $\boldsymbol{c}_{\frac{1}{f}}$ but promoted to the $(a+1, b+1, c+1)$ Jacobi basis. We let $\tilde{\cdot}$ indicate promotion *to* $(a+1, b+1, c+1)$ from any other parameter set. The above equality holds for any point $\boldsymbol{x} \in \mathcal{T}^2$. So, for a solution $\boldsymbol{u}$, the coefficient operation must be such that:

$$\left(\tilde{\mathcal{D}}_{x_1} \boldsymbol{u}\boldsymbol{u}^T \tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2} \boldsymbol{u}\boldsymbol{u}^T \tilde{\mathcal{D}}_{x_2}^T\right) = \tilde{\boldsymbol{c}}_{\frac{1}{f}} \tilde{\boldsymbol{c}}_{\frac{1}{f}}^T. \quad (1.9)$$

Now, we seek a coefficient solution $\boldsymbol{u}$ which represents the solution *globally* in $\mathcal{T}^2$. However, polynomials are differentiable, and some of the problems we are interested in yield non-differentiable solutions. This is true even in the most basic case of $f \equiv 1$, in which u is the signed distance function from $\boldsymbol{x}$ to $\partial\mathcal{T}^2$, depicted below:

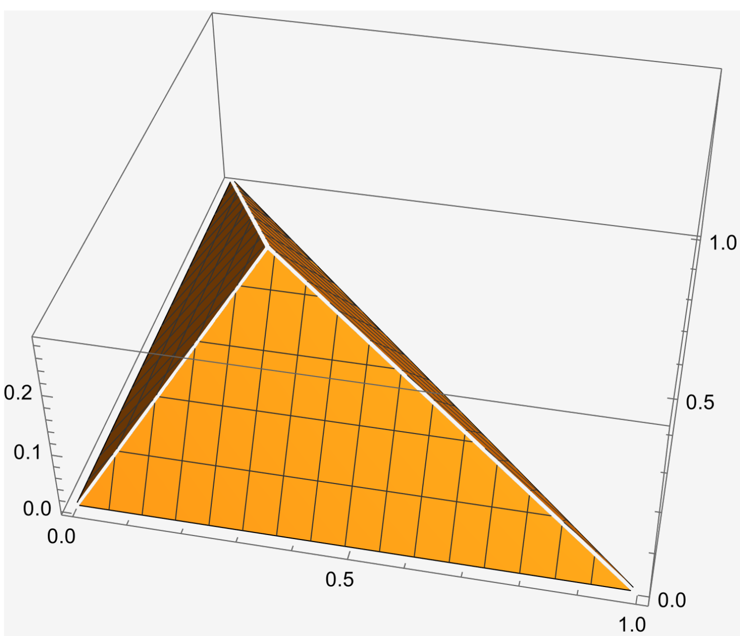

FIG. 1. *A simple calculation shows that the minimum distance from* $\boldsymbol{x} \in \mathcal{T}^2$ *to the boundary* $\partial\mathcal{T}^2$ *is given by the minimum of* $\left\{x_1, x_2, \left|\left|\boldsymbol{x} - \begin{bmatrix}(x_1 - x_2 + 1)/2 \\ (x_2 - x_1 + 1)/2\end{bmatrix}\right|\right|_2\right\}$

We cannot hope to recover a solution $\boldsymbol{u}$ which satisfies (1.9) to machine precision, as that would imply the global solution has zero residual *pointwise* everywhere. Hence, we must return to (1.7), and consider solving it weakly, in the sense of:

(1.10)

$$\int_{\mathcal{T}^2} w^{(a+1,b+1,c+1)}(\boldsymbol{x})\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T\tilde{\boldsymbol{c}}_{\frac{1}{f}}\tilde{\boldsymbol{c}}_{\frac{1}{f}}^T\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})d\boldsymbol{x} \approx ||1/f||^2_{w^{(a+1,b+1,c+1)}}$$
$$\approx \int_{\mathcal{T}^2} w^{(a+1,b+1,c+1)}(\boldsymbol{x})\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})^T\left(\tilde{\mathcal{D}}_{x_1}\boldsymbol{u}\boldsymbol{u}^T\tilde{\mathcal{D}}_{x_1}^T + \tilde{\mathcal{D}}_{x_2}\boldsymbol{u}\boldsymbol{u}^T\tilde{\mathcal{D}}_{x_2}^T\right)\boldsymbol{P}^{(a+1,b+1,c+1)}(\boldsymbol{x})d\boldsymbol{x}$$
$$\approx ||\partial_x u||^2_{w^{(a+1,b+1,c+1)}} + ||\partial_y u||^2_{w^{(a+1,b+1,c+1)}}. \qquad \blacksquare$$

Here, the norm is viewed as that of a function in $L^2(\mathcal{T}^2, w^{(a+1,b+1,c+1)})$.

Parsing this equation out a bit further (condensing sub/super scripts), we have

$$\int_{\mathcal{T}^2} \boldsymbol{P}(\boldsymbol{x})^T(\boldsymbol{u_x}\boldsymbol{u_x}^T + \boldsymbol{u_y}\boldsymbol{u_y}^T)w(\boldsymbol{x})\boldsymbol{P}(\boldsymbol{x})d\boldsymbol{x} \quad (\textbf{Only } x \textbf{ part handled below})$$
$$= \int_{\mathcal{T}^2} \boldsymbol{P}(\boldsymbol{x})^T \begin{bmatrix} u_{x,1}^2 & u_{x,1}u_{x,2} & \cdots & u_{x,1}u_{x,N} \\ u_{x,2}u_{x,1} & u_{x,2}^2 & \cdots & u_{x,2}u_{x,N} \\ \vdots & \ddots & \ddots & \\ u_{x,N}u_{x,1} & u_{x,N}u_{x,2} & \cdots & u_{x,N}^2 \end{bmatrix} \boldsymbol{P}(\boldsymbol{x})w(\boldsymbol{x})d\boldsymbol{x}$$
$$= \int_{\mathcal{T}^2} \begin{bmatrix} P_1(\boldsymbol{x}) & P_2(\boldsymbol{x}) & \cdots & P_N(\boldsymbol{x}) \end{bmatrix} \begin{bmatrix} u_{x,1}^2 & u_{x,1}u_{x,2} & \cdots & u_{x,1}u_{x,N} \\ u_{x,2}u_{x,1} & u_{x,2}^2 & \cdots & u_{x,2}u_{x,N} \\ \vdots & \ddots & \ddots & \\ u_{x,N}u_{x,1} & u_{x,N}u_{x,2} & \cdots & u_{x,N}^2 \end{bmatrix} \begin{bmatrix} P_1(\boldsymbol{x}) \\ P_2(\boldsymbol{x}) \\ \vdots \\ P_N(\boldsymbol{x}) \end{bmatrix} w(\boldsymbol{x})d\boldsymbol{x}$$
$$= \int_{\mathcal{T}^2} \begin{bmatrix} P_1(\boldsymbol{x}) & P_2(\boldsymbol{x}) & \cdots & P_N(\boldsymbol{x}) \end{bmatrix} \begin{bmatrix} u_{x,1}\sum_{j=1}^N u_{x,j}P_j(\boldsymbol{x}) \\ u_{x,2}\sum_{j=1}^N u_{x,j}P_j(\boldsymbol{x}) \\ \vdots \\ u_{x,N}\sum_{j=1}^N u_{x,j}P_j(\boldsymbol{x}) \end{bmatrix} w(\boldsymbol{x})d\boldsymbol{x}$$
$$= \int_{\mathcal{T}^2} w(\boldsymbol{x})\sum_{k=1}^N P_k(\boldsymbol{x})u_{x,k}\sum_{j=1}^N u_{x,j}P_j(\boldsymbol{x})d\boldsymbol{x} = \int_{\mathcal{T}^2} w(\boldsymbol{x})\sum_{k=1}^N\sum_{j=1}^N u_{x,k}u_{x,j}P_k(\boldsymbol{x})P_j(\boldsymbol{x})d\boldsymbol{x}$$
$$= \int_{\mathcal{T}^2}\sum_{k=1}^N u_{x,k}^2P_k^2(\boldsymbol{x})w(\boldsymbol{x})d\boldsymbol{x} = \sum_{k=1}^N u_{x,k}^2\int_{\mathcal{T}^2}P_k^2(\boldsymbol{x})dw(\boldsymbol{x}) \quad (\textbf{ now we add back in } y \textbf{ part })$$
$$= \sum_{k=1}^N u_{x,k}^2 + \sum_{k=1}^N u_{y,k}^2$$
$$= \boldsymbol{u}^T(\tilde{\mathcal{D}}_x^T\tilde{\mathcal{D}}_x + \tilde{\mathcal{D}}_y^T\tilde{\mathcal{D}}_y)\boldsymbol{u} = \tilde{\boldsymbol{c}}_{\frac{1}{f}}^T\tilde{\boldsymbol{c}}_{\frac{1}{f}}, \qquad \blacksquare$$

where we've used the mutual orthonormality relations, which persist even within a subspace of homogeneous polynomials of degree m, to eliminate all off-diagonal terms in the sum. Note, not all orthonormal bases are created equal, and we normally would have to retain many outer product terms in the sum because orthonormality between vectors of the same subspace is not a given.

Thus, we want to solve the following quadratic program with equality constraints:

$$\min_{\boldsymbol{u}\in\mathbb{R}^N} F(\boldsymbol{u}) = \boldsymbol{u}^TG\boldsymbol{u} - \tilde{\boldsymbol{c}}_{\frac{1}{f}}^T\tilde{\boldsymbol{c}}_{\frac{1}{f}} \tag{1.11}$$
$$\text{subject to} \quad \boldsymbol{u}\cdot\boldsymbol{P}(\boldsymbol{x}) = 0, \quad \boldsymbol{x}\in\partial\Omega,$$

where $G = \tilde{\mathcal{D}}_x^T \tilde{\mathcal{D}}_x + \tilde{\mathcal{D}}_y^T \tilde{\mathcal{D}}_y \in \mathbb{R}^{N \times N}$ is symmetric and positive semi-definite [1]. The gradient and Hessian of F are given by

$$\nabla_{\boldsymbol{u}} F(\boldsymbol{u}) = 2G\boldsymbol{u}, \tag{1.12}$$

$$\nabla_{\boldsymbol{u}}^2 F(\boldsymbol{u}) = 2G, \tag{1.13}$$

and we still need to consider the constraint equation.

We take the approach of eliminating the constraints by solving a the minimization problem on auxiliary variables. That is, suppose $\boldsymbol{P}(\boldsymbol{x}) \in \mathbb{R}^{M \times N}$, $M > N$, has rank p. We can compute the *full SVD* of $\boldsymbol{P}(\boldsymbol{x}) = U\Sigma V^T$. Then, the last $n-p$ rows of V^T form a basis for the $\mathcal{N}(\boldsymbol{P}(\boldsymbol{x}))$. That is, $V^T = \begin{bmatrix} V_p^T \\ V_{n-p}^T \end{bmatrix}$, and any solution $\boldsymbol{u}$ satisfying the equality constraints can be expressed as

$$\boldsymbol{u} = V_{n-p}\boldsymbol{u}_{\text{eq}}, \tag{1.14}$$

Hence, we can solve

$$\min_{\boldsymbol{u}_{\text{eq}} \in \mathbb{R}^N} F(\boldsymbol{u}_{\text{eq}}) = \boldsymbol{u}_{\text{eq}}^T V_{n-p}^T G V_{n-p} \boldsymbol{u}_{\text{eq}} - \tilde{\boldsymbol{c}}_{\frac{1}{f}}^T \tilde{\boldsymbol{c}}_{\frac{1}{f}} \tag{1.15}$$

and finally find $\boldsymbol{u}$ by computing as in (1.14). Note, the gradient and Hessian of the auxiliary problem are given by

$$\nabla_{\boldsymbol{u}_{\text{eq}}} F(\boldsymbol{u}_{\text{eq}}) = 2V_{n-p}^T G V_{n-p} \boldsymbol{u}_{\text{eq}}, \tag{1.16}$$

$$\nabla_{\boldsymbol{u}}^2 F(\boldsymbol{u}) = 2V_{n-p}^T G V_{n-p}. \tag{1.17}$$

Note, we can find p by the number of singular values above some small threshold that we choose (near $\varepsilon_{\text{mach}}$) so that it represents numerical rank.

Remark 1.1. There is a ton of material in chapter 12 and 16 of the Nocedal/Wright optimization book. I believe I can prove properties on minimizers to (1.11) (though unnecessary) and maybe use a direct method therein. There still remains the problem of initialization..perhaps we just initialize from the null space, or solve a linearized problem?

Remark 1.2. By making use of a weighted basis $\tilde{\boldsymbol{P}}(\boldsymbol{x})$, and expressing our solution coefficients and differential operators in terms of this weighted basis, we can alternatively eliminate the constraint equation and solve a completely unconstrained quadratic optimization problem I think I need to do some more symbolic computation to figure out the form of the RHS and/or additional conditions u must first satisfy so that we solve the same problem. I implemented that weighted basis, but I don't think I will get things to work properly, or rather, I will not continue trying.We provide more details in the next section.

[1] Note, $\forall \boldsymbol{x} \in \mathbb{R}_{\star}^{n^+}$, $\boldsymbol{x}^T G \boldsymbol{x} = \boldsymbol{x}^T \tilde{\mathcal{D}}_x^T \tilde{\mathcal{D}}_x \boldsymbol{x} + \boldsymbol{x}^T \tilde{\mathcal{D}}_y^T \tilde{\mathcal{D}}_y \boldsymbol{x} = ||\tilde{\mathcal{D}}_x \boldsymbol{x}||^2 + ||\tilde{\mathcal{D}}_y \boldsymbol{x}||^2 \geq 0$. We have equality only for any $\boldsymbol{x} = \alpha \boldsymbol{e}_1$, as derivatives of the constant polynomial are 0.

1.2. A variational approach. Let's try framing the problem from the perspective of infinite dimensional optimization. Thus far, my efforts have fallen under the discretize-then-optimize approach for solving non-linear PDEs. So, let's flip that and consider the optimize-then-discretize approach. The first order conditions for optimality of the functional may serve as a way to initialize the problem for the discretized non-linear optimization call.

We want to solve

$$\begin{aligned} &\min_{u \in L_2(\mathcal{T}^2, w^{(a,b,c)})} \mathcal{L}(u), \\ &\text{s.t. } u = q \text{ on } \partial\mathcal{T}^2 \end{aligned}, \tag{1.18}$$

where $\mathcal{L} : L_2(\mathcal{T}^2, w^{(a,bc)}) \to \mathbb{R}^+$ is defined by

$$\mathcal{L}(u) := \int_{\mathcal{T}^2} \Big[|\nabla_{\boldsymbol{x}} u|^2 - \big(\frac{1}{f}\big)^2 \Big]^{\frac{1}{2}} d\boldsymbol{x}, \tag{1.19}$$

and u, f are as in (1.1), restricted to $L_2(\mathcal{T}^2, \mathcal{W}^{(a,bc)})$.

The steps here are to first take the Gateaux derivative of $\mathcal{L}$ as

$$\delta\mathcal{L}(u, \hat{u}) = \partial\varepsilon\mathcal{L}(u + \varepsilon\hat{u})|_{\varepsilon=0} = 0,$$

where $\hat{u} \in \mathcal{H}_0^1(\mathcal{T}^2)$. This gives the necessary condition a minimizer of $\mathcal{L}$ must satisfy. Before proceeding, recall Green's identity for $u, v \in H^1(\mathcal{T}^2)$, i.e. a multidimensional integration by parts formula:

$$\int_{\mathcal{T}^2} \nabla_{\boldsymbol{x}} u \cdot v d(\boldsymbol{x}) = \int_{\partial\mathcal{T}^2} (uv) \cdot \boldsymbol{n} dS(\boldsymbol{x}) - \int_{\mathcal{T}^2} u \nabla_{\boldsymbol{x}} \cdot v d(\boldsymbol{x}). \tag{1.20}$$

Accordingly, we have

$$\begin{aligned} 0 &= \partial_\varepsilon \mathcal{L}(u + \varepsilon\hat{u})|_{\varepsilon=0} \\ &= \partial_\epsilon \Big[\int_{\mathcal{T}^2} \Big((u_x^2 + u_y^2) + 2\varepsilon(\hat{u}_x u_x + \hat{u}_y u_y) + \varepsilon^2(\hat{u}_x^2 + \hat{u}_y^2) - \big(\frac{1}{f}\big)^2 \Big)^{\frac{1}{2}} d(\boldsymbol{x}) \Big] \Big|_{\varepsilon=0} \\ &= \Big[\frac{1}{2} \int_{\mathcal{T}^2} \frac{\Big(2(\hat{u}_x u_x + \hat{u}_y u_y) + 2\varepsilon(\hat{u}_x^2 + \hat{u}_y^2) \Big)}{\Big((u_x^2 + u_y^2) + 2\varepsilon(\hat{u}_x u_x + \hat{u}_y u_y) + \varepsilon^2(\hat{u}_x^2 + \hat{u}_y^2) - \big(\frac{1}{f}\big)^2 \Big)^{\frac{1}{2}}} d(\boldsymbol{x}) \Big] \Big|_{\varepsilon=0}. \end{aligned}$$

The first variation is then given by

$$\delta\mathcal{L}(u, \hat{u}) = \int_{\mathcal{T}^2} \frac{\nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} \hat{u}}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - (\frac{1}{f})^2}} d(\boldsymbol{x}). \tag{1.21}$$

We can apply (1.20) with $u \leftarrow \hat{u}$ and $v \leftarrow \frac{\nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}}$ to derive the strong (PDE) form of the first order necessary condition a minimizer of $\mathcal{L}$ must satisfy. Using the fact that $\hat{u}$ is 0 on the boundary, we have

$$\begin{aligned} \delta\mathcal{L}(u, \hat{u}) &= \int_{\partial\mathcal{T}^2} \hat{u} \frac{\nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}} \cdot \boldsymbol{n} dS(\boldsymbol{x}) - \int_{\mathcal{T}^2} \hat{u} \nabla_{\boldsymbol{x}} \cdot \frac{\nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}} d(\boldsymbol{x}) \\ &= -\int_{\mathcal{T}^2} \hat{u} \nabla_{\boldsymbol{x}} \cdot \frac{\nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}} d(\boldsymbol{x}) = 0 \end{aligned} \tag{1.22}$$

Furthermore, the choice of $\hat{u}$ is arbitrary. Thus, we find that a minimizer u satisfies the following PDE in $\mathcal{T}^2$;

$$\begin{aligned} -\nabla_{\boldsymbol{x}} \cdot \left(\frac{\nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}} \right) &= 0, \quad \boldsymbol{x} \in \mathcal{T}^2, \\ u &= q, \quad \boldsymbol{x} \in \partial\mathcal{T}^2. \end{aligned} \tag{1.23}$$

Next, we compute the Frechet derivative $\delta^2\mathcal{L}(u, \hat{u}, \tilde{u})$, where $\tilde{u} \in H_0^1(\mathcal{T}^2)$. Letting $u \leftarrow u + \varepsilon\tilde{u}$ in (1.21), we have

$$\begin{aligned} \delta^2\mathcal{L}(u, \hat{u}, \tilde{u}) &= \partial_\varepsilon \delta\mathcal{L}(u + \varepsilon\tilde{u}, \hat{u})\Big|_{\varepsilon=0} \\ &= \left(\partial_\varepsilon \int_{\mathcal{T}^2} \frac{\nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} \hat{u} + \varepsilon \nabla_{\boldsymbol{x}} \tilde{u} \cdot \nabla_{\boldsymbol{x}} \hat{u}}{\sqrt{-\frac{1}{f^2} + \nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} u + 2\varepsilon \nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} \tilde{u} + \varepsilon^2 \nabla_{\boldsymbol{x}} \tilde{u} \cdot \nabla_{\boldsymbol{x}} \tilde{u}}} d(\boldsymbol{x}) \right)\Big|_{\varepsilon=0} \\ &= \int_{\mathcal{T}^2} \left(\frac{\nabla_{\boldsymbol{x}} \tilde{u} \cdot \nabla_{\boldsymbol{x}} \hat{u}}{\sqrt{-\frac{1}{f^2} + |\nabla_{\boldsymbol{x}} u|^2}} - \frac{(\nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} \hat{u})(\nabla_{\boldsymbol{x}} u \cdot \nabla_{\boldsymbol{x}} \tilde{u})}{(-\frac{1}{f^2} + |\nabla_{\boldsymbol{x}} u|^2)^{\frac{3}{2}}} \right) d(\boldsymbol{x}) \end{aligned} \tag{1.24}$$

The weak form of the Newton system at a point $u^k \in H_0^1(\mathcal{T}^2)$ is then independent of $\hat{u}$, and given by

$$\begin{aligned} &\text{Find } \tilde{u} \in H_0^1(\mathcal{T}^2) \text{ s.t.} \\ &\delta^2\mathcal{L}(u^k)(\hat{u}, \tilde{u}) = -\delta\mathcal{L}(u^k)(\hat{u}), \\ &\text{and set } u^{k+1} = u^k + \tilde{u}, \\ &\text{with } u^0 = q \text{ on } \partial\mathcal{T}^2. \end{aligned} \tag{1.25}$$

To derive the strong form of the Newton system, we proceed as for the first variation using integration by parts formula. Let $u \leftarrow \hat{u}$ and $v \leftarrow \frac{\nabla_{\boldsymbol{x}} \tilde{u}}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}}$ for the first term in (1.24). For the second term therein, let $u \leftarrow \hat{u}$ and $v \leftarrow \frac{\nabla_{\boldsymbol{x}} \tilde{u} |\nabla_{\boldsymbol{x}} u|^2}{(|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2})^{\frac{3}{2}}}$. Using the fact that $\hat{u} \in H_0^1(\mathcal{T}^2)$, and incorporating the first variation from (1.21) as needed in (1.25), we find the Newton system is equivalently expressed as

$$\int_{\mathcal{T}^2} \hat{u} \Big[\nabla_{\boldsymbol{x}} \cdot \Big(\frac{\nabla_{\boldsymbol{x}} \tilde{u} |\nabla_{\boldsymbol{x}} u|^2}{(|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2})^{\frac{3}{2}}} - \frac{\nabla_{\boldsymbol{x}} \tilde{u} + \nabla_{\boldsymbol{x}} u}{\sqrt{|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2}}} \Big) \Big] d(\boldsymbol{x}) = 0$$

Then, by the arbitrariness of $\hat{u}$, we find the strong form of the Newton system is

$$\begin{gathered} \text{Find } \tilde{u} \in H_0^1(\mathcal{T}^2) \text{ s.t.} \\ \nabla_{\boldsymbol{x}} \cdot \Big[\nabla_{\boldsymbol{x}} \tilde{u} \Big(\frac{|\nabla_{\boldsymbol{x}} u^k|^2}{(|\nabla_{\boldsymbol{x}} u^k|^2 - \frac{1}{f^2})^{\frac{3}{2}}} - \frac{1}{\sqrt{|\nabla_{\boldsymbol{x}} u^k|^2 - \frac{1}{f^2}}} \Big) \Big] = \nabla_{\boldsymbol{x}} \cdot \Big(\frac{\nabla_{\boldsymbol{x}} u^k}{\sqrt{|\nabla_{\boldsymbol{x}} u^k|^2 - \frac{1}{f^2}}} \Big) \\ \text{and set } u^{k+1} = u^k + \tilde{u}, \\ \text{with } u^0 = q \text{ on } \partial\mathcal{T}^2. \end{gathered} \tag{1.26}$$

By choosing $u^0 = q$ on $\partial\mathcal{T}^2$, and setting $u^{k+1} = u^k + \tilde{u}$, we have an infinite-dimensional newton stepper for solving the Eikonal problem. To avoid the trivial solution $u \equiv 0$ when $q \equiv 0$, we can try this with

$$u^0 = w^{(3/2,3/2,3/2)}(x, y) = xy(1 - x - y),$$

and use the weighted basis for differentiation in coefficient space. We iterate until the relative step difference is within some tolerance, in the $L_2(\mathcal{T}^2, w)$ norm. I will give details on the weighted Laplace operator momentarily.

Algorithm 1.1 Netwon updates for solving (1.1)

Require: `tol` < 1
  $u^k \leftarrow xy(1 - x - y)$
  $\tilde{u} \leftarrow 1$
  while $\left(|\nabla_{\boldsymbol{x}} u^k|^2 - (\frac{1}{f})^2 > 0\right)$ and $||\tilde{u}||_{L^2(\mathcal{T}^2,\tilde{w})} >$ `tol` **do**
    $\tilde{u} \leftarrow$ solution to (1.26)
    $u^{k+1} \leftarrow u^k + \tilde{u}$
    $u^k \leftarrow u^{k+1}$
  end while

Note, Algorithm 1.1 as well as the preceding derivations of weak and strong forms all assume that $|\nabla_{\boldsymbol{x}} u^k|^2 - (\frac{1}{f})^2 > 0$ for any k. To prevent the need for this check, we can consider the modified (though equivalent) functional

$$\mathcal{L}(u) = \int_{\mathcal{T}^2} [(|\nabla_{\boldsymbol{x}} u|^2 - \frac{1}{f^2})^2]^{\frac{1}{4}} \tag{1.27}$$

Even though (1.27) is equivalent to (1.19), the latter does not necessarily ensure that the range of $\mathcal{L}$ is $\mathbb{R}^+$ without specifying the appropriate branch of $\surd$ to consider, while the modified functional does without us having to deal with absolute values or the sgn function. Any arguments raised to the $\frac{1}{2}$ or $\frac{3}{2}$ powers, as they are in (1.26), can simply be squared and square root-ed before exponentiation. That is, the numerical implementation will include such a transformation to those arguments (this might work? we'll see). Expressing (1.26) in modal form is our next challenge.

2. The weighted Jacobi basis. The weighted basis

$$\tilde{\boldsymbol{P}}^{(a,b,c)}(x,y) = x^{a-\frac{1}{2}}y^{b-\frac{1}{2}}(1-x-y)^{c-\frac{1}{2}}\boldsymbol{P}^{(a,b,c)} \tag{2.1}$$

will prove useful for incorporating boundary conditions to the discrete operator, while preserving its sparsity. Note, we are simply multiplying the original basis by the weight function $W^{(a,b,c)}(\boldsymbol{x})$ which forms the measure under which they are orthogonal. We need the following weighted ladder operators for differentiation in the weighted basis:

$$\begin{aligned} -(2k+b+c+1)h_{k,n}^{(a,b,c)}\partial_x\tilde{P}_{k,n}^{(a,b,c)} &= (k+c)(n-k+1)h_{k,n+1}^{(a-1,b,c-1)}\tilde{P}_{k,n+1}^{(a-1,b,c-1)} \\ &+ (k+1)(n-k+a)h_{k+1,n+1}^{(a-1,b,c-1)}\tilde{P}_{k+1,n+1}^{(a-1,b,c-1)} \end{aligned} \tag{2.2}$$

$$h_{k,n}^{(a,b,c)}\partial_y\tilde{P}_{k,n}^{(a,b,c)} = -(k+1)h_{k+1,n+1}^{(a,b-1,c-1)}\tilde{P}_{k+1,n+1}^{(a,b-1,c-1)}. \tag{2.3}$$

If $\boldsymbol{f}$ are the coefficients of f under the (a,b,c) basis, then we have sparse banded matrices $\mathcal{W}_{x,(a,b,c)}^{(a-1,b,c-1)}, \mathcal{W}_{y,(a,b,c)}^{(a,b-1,c-1)}$ such that

$$\begin{aligned} \partial_x[x^{a-\frac{1}{2}}y^{b-\frac{1}{2}}(1-x-y)^{c-\frac{1}{2}}f(x,y)] &= \partial_x(\tilde{\boldsymbol{P}}^{(a,b,c)}(x,y)^T\boldsymbol{f}) \\ &= \tilde{\boldsymbol{P}}^{(a-1,b,c-1)}(x,y)^T\mathcal{W}_{x,(a,b,c)}^{(a-1,b,c-1)}\boldsymbol{f}, \end{aligned} \tag{2.4}$$

$$\begin{aligned} \partial_y[x^{a-\frac{1}{2}}y^{b-\frac{1}{2}}(1-x-y)^{c-\frac{1}{2}}f(x,y)] &= \partial_y(\tilde{\boldsymbol{P}}^{(a,b,c)}(x,y)^T\boldsymbol{f}) \\ &= \tilde{\boldsymbol{P}}^{(a,b-1,c-1)}(x,y)^T\mathcal{W}_{y,(a,b,c)}^{(a,b-1,c-1)}\boldsymbol{f} \end{aligned} \tag{2.5}$$

2.0.1. Lowering operators. Additionally, we have lowering operators which relate $xP_{k,n}^{(a,b,c)}$ to $P_{k,n}^{(a-1,b,c)}$ and $yP_{k,n}^{(a,b,c)}$ to $P_{k,n}^{(a,b-1,c)}$. These are given by the following recurrences:

$$\begin{aligned} (2n+a+b+c+2)h_{k,n}^{(a,b,c)}xP_{k,n}^{(a,b,c)} &= (n-k+a)h_{k,n}^{(a-1,b,c)}P_{k,n}^{(a-1,b,c)} && (2.6) \\ &+ (n-k+1)h_{k,n+1}^{(a-1,b,c)}P_{k,n+1}^{(a-1,b,c)}, && (2.7) \\ (2k+b+c+1)(2n+a+b+c+2)h_{k,n}^{(a,b,c)}yP_{k,n}^{(a,b,c)} &= (k+b)(n+k+b+c+1)h_{k,n}^{(a,b-1,c)}P_{k,n}^{(a,b-1,c)} \\ &- (k+1)(n-k+a)h_{k+1,n}^{(a,b-1,c)}P_{k+1,n}^{(a,b-1,c)} \\ &- (k+b)(n-k+1)h_{k,n+1}^{(a,b-1,c)}P_{k,n+1}^{(a,b-1,c)} \\ &+ (k+1)(n+k+a+b+c+2)h_{k+1,n+1}^{(a,b-1,c)}P_{k+1,n+1}^{(a,b-1,c)}. \blacksquare && (2.8) \end{aligned}$$

So, there are sparse banded matrices $\mathcal{L}_{(a,b,c)}^{(a-1,b,c)}$ and $\mathcal{L}_{(a,b,c)}^{(a,b-1,c)}$ corresponding to multiplication by x and y, respectively, such that

$$xf(x,y) = x\boldsymbol{P}^{(a,b,c)}(x,y)^T\boldsymbol{f} = \boldsymbol{P}^{(a-1,b,c)}(x,y)^T\mathcal{L}_{(a,b,c)}^{(a-1,b,c)}\boldsymbol{f}, \tag{2.9}$$

$$yf(x,y) = y\boldsymbol{P}^{(a,b,c)}(x,y)^T\boldsymbol{f} = \boldsymbol{P}^{(a,b-1,c)}(x,y)^T\mathcal{L}_{(a,b,c)}^{(a,b-1,c)}\boldsymbol{f}. \tag{2.10}$$

2.0.2. Discrete differential operators. We will construct the modal Eikonal operator which operates on coefficients under the basis $\tilde{\boldsymbol{P}}^{(3/2,3/2,3/2)}$ and produces coefficients under $\boldsymbol{P}^{(3/2,3/2,3/2)}$. Note, *the conversion from the weighted to regular*

Jacobi basis depends on our choice of parameters $a = b = c = 3/2$. With $\boldsymbol{f}$ the coefficients of $f(x, y)$ in the (a, b, c) basis,

$$\begin{aligned}
&\partial_x[W^{(a,b,c)}(x,y)f(x,y)] = \tilde{\boldsymbol{P}}^{(a-1,b,c-1)}(x,y)^T\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\boldsymbol{f}\\
&= x^{a-\frac{3}{2}}y^{b-\frac{3}{2}}(1-x-y)^{c-\frac{3}{2}}\left(y\boldsymbol{P}^{(a-1,b,c-1)}(x,y)\right)^T(\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\boldsymbol{f})\\
&= x^{a-\frac{3}{2}}y^{b-\frac{3}{2}}(1-x-y)^{c-\frac{3}{2}}\boldsymbol{P}^{(a-1,b-1,c-1)}(x,y)^T(\mathcal{L}^{(a-1,b-1,c-1)}_{(a-1,b,c-1)}\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\boldsymbol{f}),\\
&\partial_y[W^{(a,b,c)}(x,y)f(x,y)] = \tilde{\boldsymbol{P}}^{(a,b-1,c-1)}(x,y)^T\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f}\\
&= x^{a-\frac{3}{2}}y^{b-\frac{3}{2}}(1-x-y)^{c-\frac{3}{2}}\left(x\boldsymbol{P}^{(a,b-1,c-1)}(x,y)\right)^T(\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f})\\
&= x^{a-\frac{3}{2}}y^{b-\frac{3}{2}}(1-x-y)^{c-\frac{3}{2}}\boldsymbol{P}^{(a-1,b-1,c-1)}(x,y)^T(\mathcal{L}^{(a-1,b-1,c-1)}_{(a,b-1,c-1)}\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f})
\end{aligned}$$

When $a = b = c = 3/2$, we have

$$\begin{aligned}
\partial_x[W^{(a,b,c)}(x,y)f(x,y)] &= \boldsymbol{P}^{(a-1,b-1,c-1)}(x,y)^T(\mathcal{L}^{(a-1,b-1,c-1)}_{(a-1,b,c-1)}\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\boldsymbol{f}),\\
\partial_y[W^{(a,b,c)}(x,y)f(x,y)] &= \boldsymbol{P}^{(a-1,b-1,c-1)}(x,y)^T(\mathcal{L}^{(a-1,b-1,c-1)}_{(a,b-1,c-1)}\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f}),
\end{aligned}$$

We need to see what auxiliary problems we must solve, as the nonlinearity of (1.1) leads to cross-terms that must cancel if we hope that a solution to the weighted problem also solves the unweighted problem. In the bulk, we have

$$[\partial_x(Wu)]^2 + [\partial_y(Wu)]^2 = W^2[u_x^2 + u_y^2] + u^2(W_x + W_y) + 2uW(W_xu_x + W_yu_y),$$

and we need the second two terms in the equation to cancel. There's something here we can do by integrating, and finding the additional conditions u must satisfy. I think it would become a solution for the initial guess of u. Essentially, we know $W^2[u_x^2 + u_y^2] = W^2/f^2$, since we assume that u solve (1.1). Then, we can enforce that

$$\int_{\mathcal{T}^2} u^2(W_x + W_y) + 2uW(W_xu_x + W_yu_y) = 0.$$

There may be some simplification that occurs if we plug in the Jacobi expansion of u .. idk

In the weighted basis, second partials acting on function modes can be constructed with the promotion and unweighted ladder operator as done in Section 2.0.2.

(2.11)
$$\partial_{xx}f(x,y) = \boldsymbol{P}^{(a,b,c)}(x,y)^T\left(\mathcal{K}^{(a,b,c)}_{(a,b-1,c)}\mathcal{D}^{(a,b-1,c)}_{x,(a-1,b-1,c-1)}\mathcal{L}^{(a-1,b-1,c-1)}_{(a-1,b,c-1)}\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\boldsymbol{f}\right),$$

(2.12)
$$\partial_{yy}f(x,y) = \boldsymbol{P}^{(a,b,c)}(x,y)^T\left(\mathcal{K}^{(a,b,c)}_{(a-1,b,c)}\mathcal{D}^{(a-1,b,c)}_{y,(a-1,b-1,c-1)}\mathcal{L}^{(a-1,b-1,c-1)}_{(a,b-1,c-1)}\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f}\right),$$

(2.13)
$$\partial_{xy}f(x,y) = \boldsymbol{P}^{(a,b,c)}(x,y)^T\left(\mathcal{K}^{(a,b,c)}_{(a,b-1,c)}\mathcal{D}^{(a,b-1,c)}_{x,(a-1,b-1,c-1)}\mathcal{L}^{(a-1,b-1,c-1)}_{(a,b-1,c-1)}\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}\boldsymbol{f}\right)$$

Δ_W, the weighted Laplace operator, is then given by

$$\begin{aligned}
\Delta_W = \;&\mathcal{K}^{(a,b,c)}_{(a,b-1,c)}\mathcal{D}^{(a,b-1,c)}_{x,(a-1,b-1,c-1)}\mathcal{L}^{(a-1,b-1,c-1)}_{(a-1,b,c-1)}\mathcal{W}^{(a-1,b,c-1)}_{x,(a,b,c)}\\
&+\mathcal{K}^{(a,b,c)}_{(a-1,b,c)}\mathcal{D}^{(a-1,b,c)}_{y,(a-1,b-1,c-1)}\mathcal{L}^{(a-1,b-1,c-1)}_{(a,b-1,c-1)}\mathcal{W}^{(a,b-1,c-1)}_{y,(a,b,c)}.
\end{aligned} \tag{2.14}$$

The cross partial terms are needed only when we include affine transformations from general triangles to the standard reference triangle, as they appear when chaining out derivatives of the parametrization.